MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548021

(REGISTER B, RANGE)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range register B, used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10548021	Register B, Range

PACKING DATA SHEET

10548021	Packaging of Range Register B

FSC 1285

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548021 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirements of 3.5.1 through 3.5.3.7 the following conditions shall apply:

a. Logic 0: 0.0 ±0.5 volt direct current (Vdc) level.

b. Logic 1: +3.5 ±1.0 Vdc level.

c. Positive pulse: a change in direct current (dc) level from 0.0 ±0.5 volt (V) to +3.5 ±1.0 V and return to 0 V.

d. Negative pulse: a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.

3.5.1 Input power. - The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals P1-24, 25 and 26 (+5 V regulated) shall be 5.0 ±0.3 Vdc; at 250 milliamperes (mA), regulated, with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signal. - The input signal shall be a positive pulse of 4 kilohertz (kHz) and 0.8 ±0.1 microsecond (μs) duration. The input signal shall be applied to the input terminals as specified in 3.5.3.1 through 3.5.3.7.

3.5.3 Output signals.

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3.5.3.1 Up count signals.- With the input signal as specified in 3.5.2 and the input logic levels applied as specified in Table I, Step 1, the output signals shall be symmetrical square waves for the set and reset side, with frequencies as specified in Table II. Comparing two consecutive set or reset stage output signals they shall be phase shifted.

TABLE I. - INPUT LOGIC LEVELS

STEP	INPUT TERMINALS		
	47	14	3
1	1	2	1
2	2	2	2
3	2	1	2

TABLE II. - OUTPUT SIGNALS

SET SIDE		RESET SIDE		
TERMINALS	(TP)	TERMINALS	(TP)	SIGNAL FREQUENCY
42	1	49	2	2 kHz
51	3	46	4	1 kHz
56	Note	54	5	500 Hz

Note: Terminal 56 connected to terminal 7 for test only. The signal is present at TP6 on the test fixture.

3.5.3.2 Down count signals. - With the input signal as specified in 3.5.2 and the input logic levels applied as specified in Table I, Step 2, the output signals shall be symmetrical square waves for the set and reset side, with frequencies as specified in Table II. Comparing two consecutive set or reset stage output signals they shall be in phase.

3.5.3.3 Counter freeze. - With the input logic levels applied as specified in Table I, Step 3, the output signals shall be as specified in Table III.

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TABLE III. - FREEZE OUTPUT SIGNALS

TERMINALS	(TP)	SIGNAL CHARACTERISTIC
42	1	Logic level (complement of 49)
49	2	Logic level (complement of 42)
51	3	Logic level (complement of 46)
46	4	Logic level (complement of 51)
54	5	Logic level (complement of 56)
56	6	Logic level (complement of 54)

3.5.3.4 Count control output signals. - With the input logic levels applied as specified in Table I, Steps 1 and 3, the count control output signals shall have the logic levels as specified in Table IV, Steps 1 and 2 respectively.

TABLE IV. - COUNT CONTROL OUTPUT SIGNALS

STEP	TERMINAL	(TP)	LEVEL	TERMINAL	(TP)	LEVEL
1	16	8	0	53	9	1
2	16	8	1	53	9	0

3.5.3.5 Pulse output. - With the input signal as specified in 3.5.2 and the input logic levels as specified in Table I, Step 2, the pulse output signal at terminal P1-8 shall be 0.8 ±0.1 μs wide, with a 500 hertz (Hz) pulse repetition frequency (PRF).

3.5.3.6 D/A gating signals. - With a logic 1 applied to terminal 47 and the input signals sequentially applied as specified in Table V, the D/A gating output signals shall be as specified in Table V. With a logic 0 applied to terminal 47 the D/A gating output signal shall be as follows:

Logic 0: terminals 9, 11, 18, 17, 2, 4, 5 and 33.
Logic 1: terminals 15 and 6.

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TABLE V. - D/A GATING SIGNALS

STEP	INPUT TERMINAL	SIGNAL SW. POS.	CHARACTERISTIC	TERMINAL	OUTPUT (TP)	SIGNAL CHARACTERISTIC
1	43	1	1	9	11	3
2	13	2	1	11	12	1
3	12	3	1	18	13	1
4	10	4	1	17	14	1
5	21	5	1	2	15	1
6	19	6	1	4	16	1
7	35	7	1	15	17	4
8	43	8	2	5	18	2
9	40	9	1	6	19	4
10	45	10	1	33	20	1

SIGNAL CHARACTERISTICS:

1. As specified in 3.5.2.
2. Positive pulse, 0.8 ±0.1 μs wide with 500 Hz PRF.
3. 500 Hz square wave.
4. Negative, pulse, 0.8 ±0.1 μs wide with 4 kHz PRF.

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3.5.3.7 Mid-range signal. - With the signals applied as specified in Table VI the mid-range output signal at terminal P1-1 shall be a negative pulse, 0.8 ±0.1 μs wide with a 500 Hz PRF.

TABLE VI. - MID-RANGE SIGNALS

TERMINAL	SIGNAL CHARACTERISTIC
48	2 kHz square wave
22	1 kHz square wave
23	500 kHz square wave
36	Positive pulse, 0.8 ±0.1 μs wide with 500 Hz PRF.
19	Positive pulse, 0.8 ±0.1 μs wide with 4 kHz PRF.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table VII shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE VII - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2.	Pulse generator	20 kHz positive pulse; 3.0 ±1.0 Vdc amplitude, 0.8 ±0.1 μs wide. Pulse rise and fall times less than 0.1 μs.
3.	Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less, sweep period 50 ns per division or greater, input resistance greater than 8 megohms; input capacitance less than 25 pF.
4.	Test fixture	As specified in Figure 1. See Table VIII for test fixture parts list.

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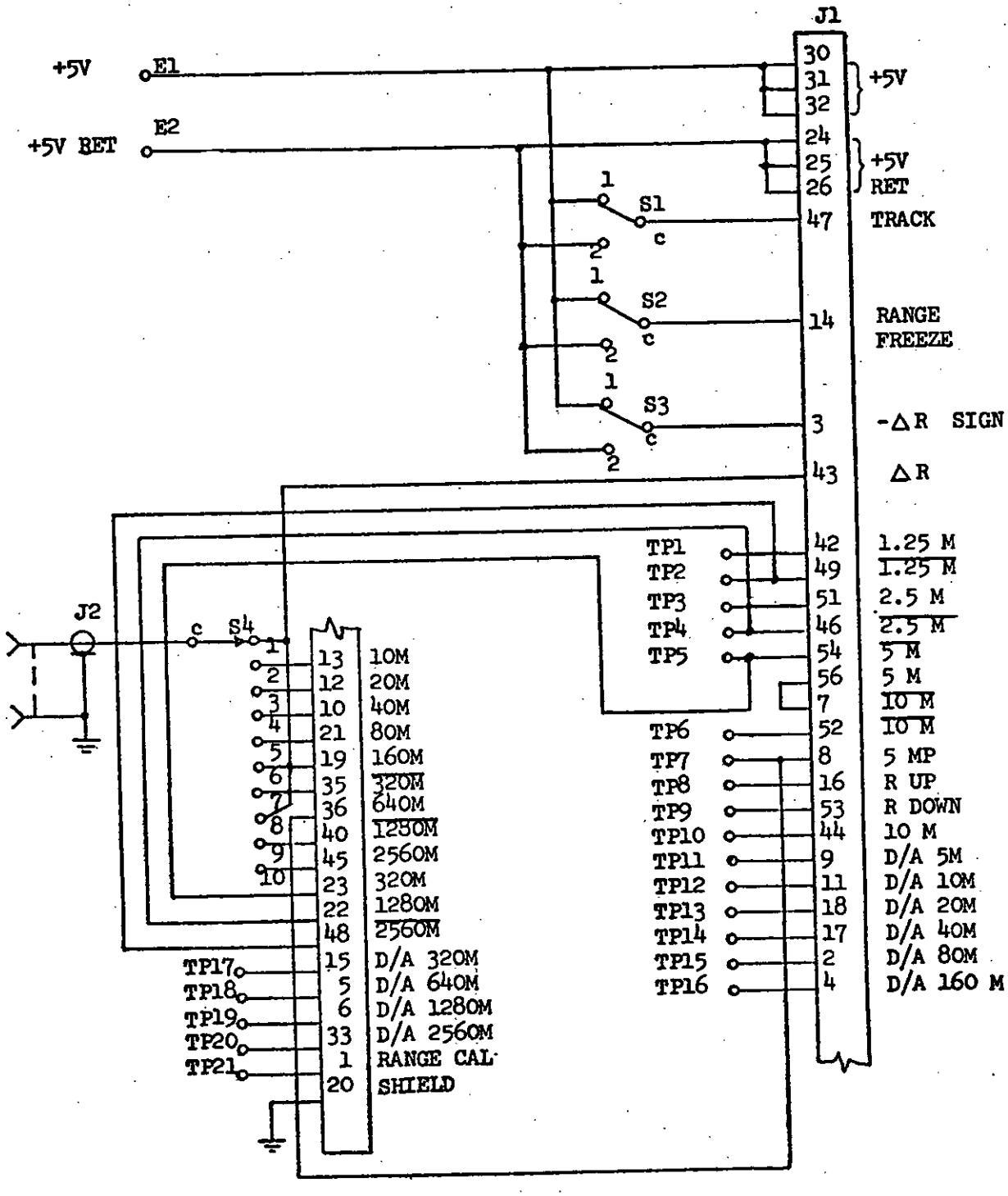


FIGURE 1. TEST FIXTURE SETUP.

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TABLE VIII - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent:
	(1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois.
	(2) 500946-01, National Connector Corp., Minneapolis, Minn.
	(3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
S1	Switch SPDT.
S2	Same as S1.
S3	Same as S2.
S4	Switch, single pole, 10 positions.
TP1 thru TP21	Tip jack, H. H. Smith 1501 or equivalent.

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4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination to determine compliance with 3.4.

4.6.3 Performance. - Set up equipment as shown in Figure 2. Use oscilloscope, item 3, for observation of input and output signals as specified in 4.6.3.1 through 4.6.3.3.7. For all signals the conditions of 3.5 shall apply. Use switches S1 through S3 to apply logic levels to terminals P1-47, P1-14, and P1-3 respectively. Switches in position 1 supply a logic 1 and in position 2 a logic 0. Use switch S4 to apply the input signal specified in 3.5.2 to the input terminals. Correlation between switch positions and signal input terminal is specified in Table V.

4.6.3.1 Input power. - Using item 1, apply input power as specified in 3.5.1.

4.6.3.2 Input signal. - Using item 2, apply input signal as specified in 3.5.2 to J2 on test fixture, item 4.

4.6.3.3 Output signals.

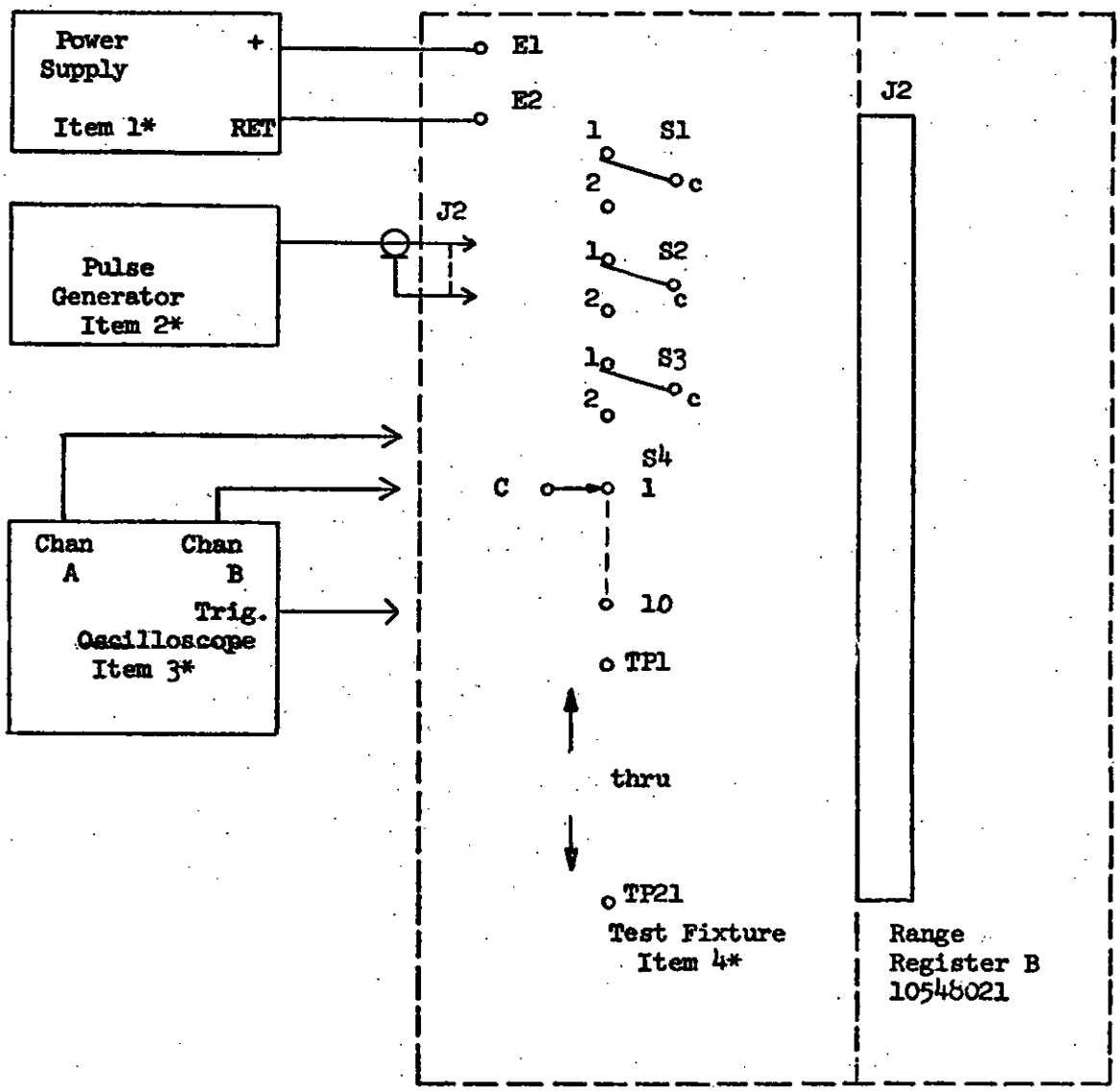
4.6.3.3.1 Up count signals. - Apply the input signal and levels as specified in 3.5.3.1. Using the test points on the test fixture, item 4, observe the output signals as specified in 3.5.3.1.

4.6.3.3.2 Down count signals. - Apply the input signals and levels as specified in 3.5.3.2. Using the test points on the test fixture, item 4, observe the output signals as specified in 3.5.3.2.

4.6.3.3.3 Counter freeze. - Apply the input logic levels as specified in 3.5.3.3. At the test points on the test fixture, item 4, specified in Table III, observe the output signals as specified in Table III. Terminal P1-56 is connected to P1-7 in the test fixture for test purposes only and the output signal is observed at TP-6 (terminal P1-52).

4.6.3.3.4 Count control output signals. - Apply the input logic levels as specified in 3.5.3.4. At the test points on the test fixture, item 4, specified in Table IV, observe the output levels as specified in Table IV.

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*Refer to 4.5.3

FIGURE 2. RANGE REGISTER B TEST SETUP.

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4.6.3.3.5 Pulse output. - Apply the input signal and levels as specified in 3.5.3.5 and at test point 7 on the test fixture, observe the output signal as specified in 3.5.3.5.

4.6.3.3.6 D/A gating signals. - Using switch S1 to apply a logic 1 level to P1-47 and switch S4 to apply the input signal as specified in Table V, observe the D/A gating output signals as specified in Table V. Use switch S1 to apply a logic 0 level to P1-47 and observe the D/A gating output signal as specified in 3.5.3.6. Use Table V for correlation between terminals and test points to observe the signals.

4.6.3.3.7 Mid-range signal. - Place switch S4 in position 1 and at test point 21 observe the mid-range output signal as specified in 3.5.3.7. The input signals as specified in 3.5.3.7 and Table VI are applied by interconnections provided on the test fixture.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548021. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The range register B is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The range register comprises the three least significant bits of the up/down counter used for storage of incremental range information. The range register B assembly also contains track/search gating logic for the range word.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A244

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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